COM-480 – Milestone 2

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General Description

Our website invites users on a journey through the life and death cycles of video games, exploring how certain titles appear to transcend these patterns. The story line is structured around three core components.

First, a main visualization offers users a broad overview of the landscape, enabling them to observe and explore the dataset as a whole. Second, an interactive sandbox clustering tool allows users to highlight games of interest, group them according to their own criteria, and draw comparisons across titles. Third, a curated static analysis presents our own findings and interpretations while deliberately leaving room for the user to challenge, extend, or deepen the conclusions we propose.

Main Visualization

The main visualization, sketched in Fig 1, serves as the primary entry point into the website's story line structure. It provides users with a dynamic view of how the gaming landscape evolves over time, and simultaneously acts as a gateway to the secondary visualization allowing users to identify games of interest and carry them forward for deeper exploration.

The visualization is defined by the following parameters:

- **Time Window:** A temporal scope that frames the data displayed.
- **X-Axis:** A value proportional to each game's release date relative to the selected time window, where games positioned furthest to the left are the most distant from their release date.
- **Y-Axis:** The average number of monthly active players, serving as a proxy for a game's sustained engagement and longevity.

The tools needed to implement this visualization are **D3.js v7** with the Margin Convention. Scales used include `d3.scaleLinear` (year), `d3.scaleLog().clamp(true)` (average players), `d3.scaleSqrt` (bubble radius), and a diverging `d3.scaleLinear` with a 3-stop domain `[0, 0.1, 0.3]` mapping alive ratio to red/amber/green. Upstream data wrangling is handled in **Python** (`pandas`). The relevant lectures are **11.1** (Bubble plot idioms), **5.2** (D3 quantitative scales and axes), and **1.1** (Visualization pipeline).

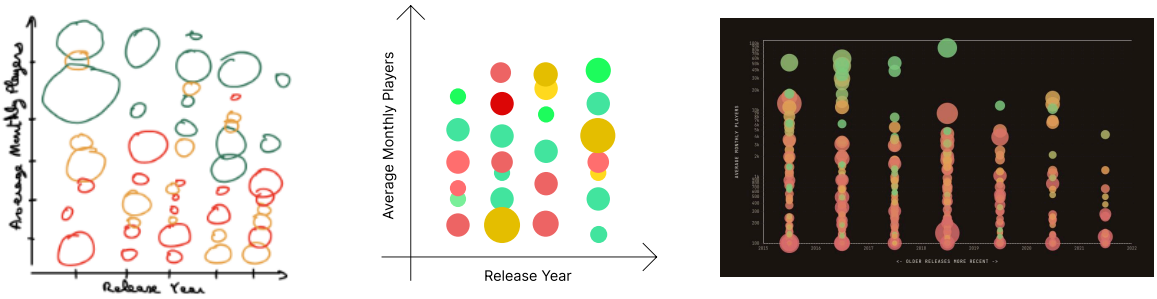

Figure 1: Sketch of the Main visualization

Downstream Visualizations 1: Game Details

Within the main visualization, individual games are represented as interactive bubbles. Each bubble is clickable, triggering the display of a dedicated identity card that surfaces key information and statistics about the selected title. This includes, but is not limited to, the game's name, release date, and genre. The sketch of the visualization can be seen in Fig 2.

Figure 2: Sketch of the Downstream visualization 1: Game Details

Game details are surfaced via `d3.select(...).on("click")` on each bubble, triggering an HTML side-panel card populated through direct `.text()` injection and toggled via a `.hidden` CSS class. A `#send-to-sandbox` button adds the selected game to `state.selectedIds`, maintained as a JS Set. The relevant lectures are **5.1** (Details on Demand; Info-Seeking Mantra) and **5.2** (D3 event handlers on selections).

Downstream Visualizations 2: Filtering

To enhance the analytical depth of the main visualization, a filtering widget, shown in Fig 3, allows users to refine the displayed games according to specific criteria, such as Game Genre, Rating and Platform(s).

By narrowing the scope of the games on display, this widget expands the expressive power of the core visualization and grants users greater freedom to tailor their exploration before transitioning to the secondary visualization.

The filtering widget is implemented using a vanilla JS state object tracking `yearRange`, `minRating`, and `genres/platforms` as Sets, with a single `getFilteredGames()` predicate applied on each update. Genre and platform chips are rendered via D3 `.data().join('button')` with `.on('click')` toggles; the rating slider is handled through an HTML `<input type="range">` element. Crossfilter is not used, as the data volume does not require it. The relevant lecture is **5.1** (Dynamic Queries; faceted filtering).

Downstream Visualizations 3: Time Window Slider

This widget, shown in Fig 3, allows to move the time window considered for the main visualization.

The time window slider is implemented using two HTML `<input type="range">` elements (`#time-slider-start` and `#time-slider-end`) forming a dual-handle slider, with handles swapped automatically if crossed. Each input event triggers `update()`, which re-runs the filter predicate and applies `d3.transition().duration(500)` to the chart's scales and bubble positions. The relevant lectures are **5.1** (Navigation via Item Reduction; Animated Transitions) and **5.2** (D3 transitions).

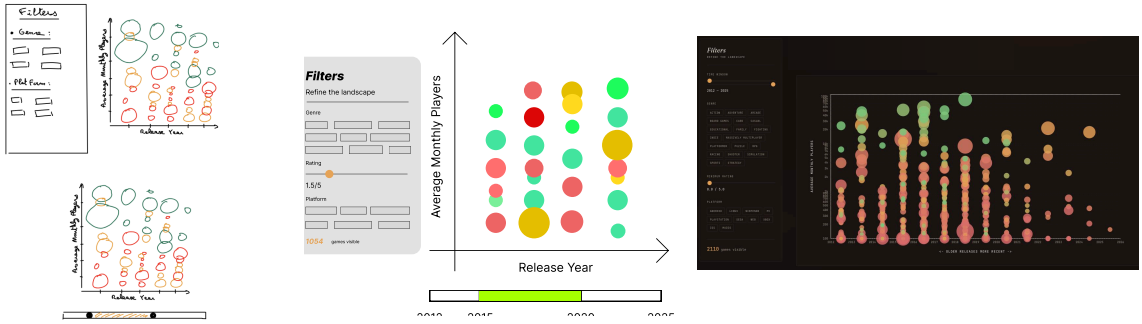

Figure 3: Sketch of the Downstream Visualizations 2: Filtering + Downstream Visualizations 3: Time Window Slider

Sandbox Visualization

The secondary visualization is designed to receive games selected by the user from the main visualization, surfacing them within a clustering graph built around multiple criteria decided by the user. In keeping with the website's overarching story line, this component empowers users to pursue their own lines of inquiry and conduct personalized analysis.

The deliberate "sandbox" character of this visualization is intended to foster deeper user engagement, encouraging exploration and interpretation rather than passive consumption of pre-defined results.

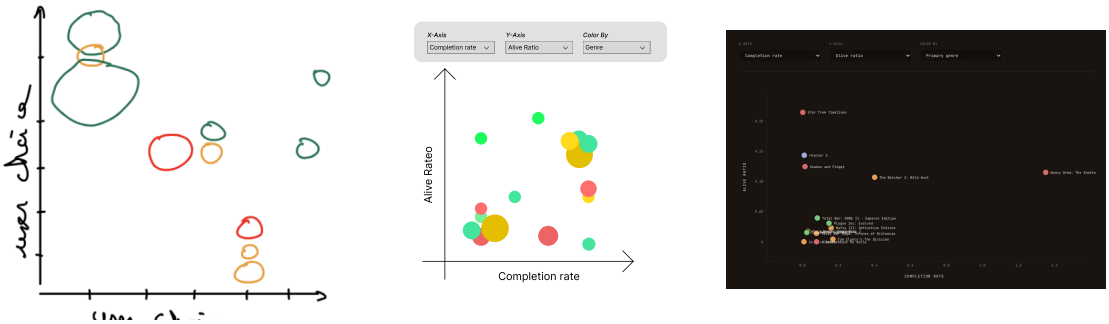

Figure 4: Sketch of the Sandbox

The sandbox visualization, shown in Fig 4, is implemented as a standalone D3 scatter view, initialized lazily via `requestAnimationFrame` once the view becomes visible. Three HTML `<select>` pickers drive `sandbox.xField/yField/colorField`; a custom `paddedDomain()` function handles edge cases where $n = 1$ or all values are equal. Points re-render with D3 transitions on each field change. Linking with the main visualization is carried through `state.selectedIds`. The relevant lectures are **10** (Multi-view scatter comparisons), **5.1** (Linking and Brushing across coordinated views), and **5.2** (D3 scales and transitions).

An additional component of the sandbox is a time series visualization displaying the number of monthly active players over time for each game currently in `state.selectedIds`. This allows users to directly compare the engagement trajectories of their selected titles, observing patterns of growth, decline, or sustained longevity side by side.

Static Related Analysis

The third and final component of the website presents our own analysis of the dataset, accompanied by the corresponding results and interpretations. The visualizations in this section are

static and time series-based, offering side-by-side comparisons of per-game metrics such as alive ratio, number of active players, and related indicators.

A central emphasis of this section is that the findings presented here are not intended to be definitive. Rather, they serve as a structured conclusion to the website's narrative one that is deliberately designed to invite the user to revisit, challenge, or extend our interpretations, whether through the core and secondary visualizations or through additional avenues outlined in the extra ideas section.

In this sense, the static analysis functions both as a closing argument and as a point of departure for further personal exploration.

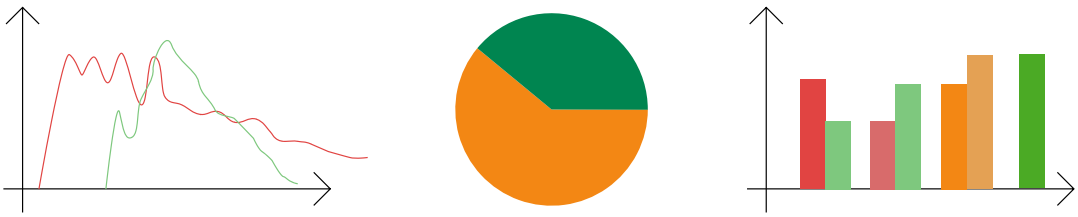

Figure 5: Sketch of Static Analysis

The whole narrative design is structured as a single-page HTML/CSS SPA with three `.view` panes (`main`, `sandbox`, `analysis`), routed by `switchView()` and enforcing a Martini Glass funnel: a guided overview, a reader-driven sandbox, and an author-driven analysis. The editorial design adopts a dark theme with Fraunces and JetBrains Mono typefaces, numbered sections, and Active Titles. No scroll-driven library is used. The relevant lectures are **12.1** (Martini Glass; author- vs reader-driven narratives; Active Titles) and **7.1** (The 4 Nested Levels of visualization design).

List of Extras Ideas

- The Search Bar: allows users to look up any specific game and access its individual statistics and dedicated visualizations directly. The primary technical challenge of this feature lies in efficiently managing computations across a sizeable game catalogue, ensuring responsive and accurate results at scale. From a narrative standpoint, the search bar is for now conceived as a complementary tool that sits alongside the main and secondary visualizations enriching the user's investigative toolkit without disrupting the coherence of the website's storyline.